

Random Variables

A **random variable** is a numerical outcome of a random experiment. It assigns numerical values to each possible outcome of a probabilistic event.

There are two main types of random variables:

1. **Discrete Random Variables**
2. **Continuous Random Variables**

1. Discrete Random Variables

A **discrete random variable** takes on a **countable number** of distinct values. These values can be finite or countably infinite.

Example 1: Rolling a Dice

Let X be a random variable representing the outcome when rolling a fair six-sided die. Possible values of X are:

$$X = 1, 2, 3, 4, 5, 6$$

Here, X is a discrete random variable because the possible outcomes are countable.

Example 2: Number of Heads in 3 Coin Tosses

Let Y be the number of heads obtained when flipping a fair coin three times. Possible values of Y are:

$$Y = 0, 1, 2, 3$$

This is discrete because it takes only specific integer values.

2. Continuous Random Variables

A **continuous random variable** can take **any value within a given range** (finite or infinite). These values are uncountable and usually represent measurements.

Example 1: Height of Students

Let H be the height of a randomly selected student in centimeters. Since height can be any real number within a range (e.g., 140.5 cm, 150.2 cm, etc.), H is a continuous random variable.

Example 2: Time Taken to Complete a Task

Let T be the time (in seconds) taken by a student to solve a math problem. Since time can take any value (e.g., 10.1s, 10.25s, 10.35s, etc.), T is continuous.

Comparison Between Discrete and Continuous Random Variables

Feature	Discrete Random Variable	Continuous Random Variable
Possible values	Countable (finite or infinite)	Uncountable (infinite range)
Examples	Number of students in a class, number of goals in a football match	Height, weight, time, temperature
Probability Calculation	Uses probability mass function (PMF)	Uses probability density function (PDF)

Probability Distributions

A **probability distribution** describes how probabilities are assigned to different possible outcomes of a random variable. It shows the likelihood of each possible value occurring.

Probability distributions are categorized based on whether the random variable is **discrete** or **continuous**:

1. Discrete Probability Distributions

For a **discrete random variable**, a probability distribution assigns probabilities to each possible value. This is described by a **Probability Mass Function (PMF)**.

Example 1: Rolling a Fair Die

Let X be the outcome of rolling a fair six-sided die.

The probability distribution is:

$$P(X = x) = \frac{1}{6}, \quad \text{for } x = 1, 2, 3, 4, 5, 6.$$

Here, each outcome has an equal probability of occurring.

Example 2: Binomial Distribution (Coin Toss)

A **binomial distribution** models the number of successes in a fixed number of independent trials. For example, if a fair coin is tossed 3 times, and X represents the number of heads, the probability distribution is:

X (Number of Heads)	Probability $P(X)$
0 (TTT)	$\frac{1}{8}$
1 (HTT, THT, TTH)	$\frac{3}{8}$
2 (HHT, HTH, THH)	$\frac{3}{8}$

X (Number of Heads)	Probability $P(X)$
3 (HHH)	$\frac{1}{8}$

This follows a **Binomial distribution** with parameters $n = 3$ (number of trials) and $p = 0.5$ (probability of heads).

2. Continuous Probability Distributions

For a **continuous random variable**, probabilities are assigned over an interval using a **Probability Density Function (PDF)**.

Example 1: Normal Distribution (Bell Curve)

The **normal distribution** is the most common continuous probability distribution. It is characterized by:

- Mean μ (center of the distribution)
- Standard deviation σ (spread of the distribution)
- A symmetric bell-shaped curve

The probability density function (PDF) of a normal distribution is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Example:

- The heights of students in a school often follow a normal distribution, where most students have heights close to the average, and fewer students have extreme heights.

Example 2: Exponential Distribution (Waiting Time)

The **exponential distribution** models the time between events in a Poisson process (e.g., time between customer arrivals at a store).

- The PDF is given by:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

Example:

- The time between consecutive phone calls at a call center follows an exponential distribution.

Comparison of Discrete and Continuous Probability Distributions

Feature	Discrete Probability Distribution	Continuous Probability Distribution
Definition	Probability assigned to distinct values	Probability assigned over an interval
Function Used	Probability Mass Function (PMF)	Probability Density Function (PDF)

Feature	Discrete Probability Distribution	Continuous Probability Distribution
Example	Binomial distribution, Poisson distribution	Normal distribution, Exponential distribution
Probability Calculation	$P(X = x)$ (exact value)	$P(a \leq X \leq b) = \int_a^b f(x)dx$ (area under curve)

Probability Mass Function (PMF)

A **Probability Mass Function (PMF)** is a function that gives the probability of a **discrete random variable** taking on a specific value. It is used for **discrete probability distributions**.

Definition

For a discrete random variable X , the probability mass function $P(X = x)$ is defined as:

$$P(X = x) = P_x$$

where:

- X is a discrete random variable,
- x represents a specific possible value of X ,
- $P(X = x)$ gives the probability that X takes the value x .

The PMF satisfies the following conditions:

1. $0 \leq P(X = x) \leq 1$ (probability is always between 0 and 1).
2. The sum of all probabilities equals 1:

$$\sum P(X = x) = 1$$

Example 1: Rolling a Fair Die

Let X be the outcome when rolling a fair six-sided die.

Since each face (1 to 6) is equally likely, the PMF is:

$$P(X = x) = \frac{1}{6}, \quad x \in \{1, 2, 3, 4, 5, 6\}$$

Here, each outcome (1, 2, 3, 4, 5, or 6) has a probability of $\frac{1}{6}$.

Example 2: Tossing a Fair Coin

Let X be the number of heads in one coin flip.

Possible values: $X = 0$ (Tails), $X = 1$ (Heads).

Since the coin is fair:

$$P(X = 0) = \frac{1}{2}, \quad P(X = 1) = \frac{1}{2}$$

Example 3: Binomial Distribution PMF

A binomial distribution represents the number of successes in n independent trials, each with a success probability p .

The PMF of a binomial distribution is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where:

- n = number of trials,
- k = number of successes,
- p = probability of success in one trial,
- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient.

Example: Flipping a Fair Coin 3 Times

Let X be the number of heads in 3 flips, with $p = 0.5$.

The PMF is:

X (Number of Heads)	$P(X)$
0 (TTT)	$\binom{3}{0} (0.5)^0 (0.5)^3 = \frac{1}{8}$
1 (HTT, THT, TTH)	$\binom{3}{1} (0.5)^1 (0.5)^2 = \frac{3}{8}$
2 (HHT, HTH, THH)	$\binom{3}{2} (0.5)^2 (0.5)^1 = \frac{3}{8}$
3 (HHH)	$\binom{3}{3} (0.5)^3 (0.5)^0 = \frac{1}{8}$

Total probability sum:

$$\frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$$

Key Properties of PMF

1. **Probability is Non-Negative:** $P(X = x) \geq 0$.
2. **Total Probability is 1:** $\sum P(X = x) = 1$.
3. **Used for Discrete Random Variables** (not for continuous ones).

Probability Density Function (PDF)

A **Probability Density Function (PDF)** describes the likelihood of a **continuous random variable** taking a particular value within a given range. Unlike a **Probability Mass Function (PMF)** (which is used for discrete variables), a **PDF does not give the probability of a single exact value** but rather describes the probability over an interval.

Definition

For a continuous random variable X , the probability density function $f(x)$ satisfies the following properties:

1. **Non-Negativity:**

$$f(x) \geq 0 \text{ for all } x.$$

2. **Total Probability Equals 1:**

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

3. **Probability Calculation Using an Interval:**

Since $P(X = x) = 0$ for a continuous variable, probability is obtained by integrating the PDF over an interval:

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

Example 1: Uniform Distribution

A **Uniform Distribution** is when all values in a range are equally likely. If X follows a **uniform distribution** over the interval $[a, b]$, its PDF is:

$$f(x) = \begin{cases} 1/(b-a), & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

Example: A Random Number Between 0 and 10

If X is a random number between 0 and 10, the PDF is:

$$f(x) = \begin{cases} 1/(10 - 0) = (1/10), & 0 \leq x \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

To find $P(2 \leq X \leq 5)$, we integrate:

$$P(2 \leq X \leq 5) = \int_2^5 \frac{1}{10} dx = \frac{5-2}{10} = \frac{3}{10}.$$

Example 2: Normal (Gaussian) Distribution

The **Normal Distribution** (or **Gaussian Distribution**) is the most common continuous probability distribution. It is characterized by a **bell-shaped curve**. The PDF of a normal distribution is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where:

- μ = mean (center of the distribution),
- σ = standard deviation (spread of the distribution).

Example: Human Heights

If the heights of students in a school follow a normal distribution with $\mu = 160 \text{ cm}$ and $\sigma = 10 \text{ cm}$, then the probability of a student having a height between 150 cm and 170 cm is:

$$P(150 \leq X \leq 170) = \int_{150}^{170} f(x)dx.$$

This requires integration or using a **normal probability table** (Z-table).

Example 3: Exponential Distribution (Waiting Time)

The **Exponential Distribution** is used to model **waiting times** (e.g., time between customer arrivals at a store). The PDF is:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

where λ is the rate parameter.

Example: Waiting Time for a Bus

If buses arrive every 10 minutes on average, the arrival time X follows an **exponential distribution** with $\lambda = \frac{1}{10}$.

The probability that a passenger waits **less than 5 minutes** is:

$$P(X \leq 5) = \int_0^5 \frac{1}{10} e^{-x/10} dx.$$

This can be computed using integration or an exponential probability table.

Comparison: PMF vs. PDF

Feature	Probability Mass Function (PMF)	Probability Density Function (PDF)
Used for	Discrete random variables	Continuous random variables
Function Type	Assigns probability to each discrete value	Defines probability over an interval
Probability Calculation	$P(X = x)$ gives probability for a specific x	$P(a \leq X \leq b) = \int_a^b f(x)dx$
Example	Rolling a die, Coin toss, Binomial distribution	Human heights, Time between events, Normal distribution

Key Takeaways

- A **PDF** is used for continuous random variables, while a **PMF** is for discrete ones.
- The **probability of an exact value in a continuous distribution is always 0**.
- Probabilities in a PDF are calculated over **intervals** by integration.
- The **Normal Distribution** and **Exponential Distribution** are common examples of PDFs.

Mathematical Expectation (Expected Value)

The **Mathematical Expectation**, also known as the **Expected Value ($E[X]$)**, represents the **average or mean value** of a random variable over many trials. It provides a measure of the central tendency of a probability distribution.

1. Definition of Expectation

For a random variable X , the expected value is denoted as:

$$E[X]$$

It is calculated differently for **discrete** and **continuous** random variables.

(A) Expected Value for a Discrete Random Variable

If X is a discrete random variable with probability mass function (PMF) $P(X = x)$, the expected value is:

$$E[X] = \sum_i x_i P(X = x_i)$$

where:

- x_i represents the possible values of X ,
- $P(X = x_i)$ is the probability of X taking the value x_i .

Example: Rolling a Fair Die

Let X be the outcome of rolling a six-sided die. The possible values of X are 1, 2, 3, 4, 5, 6, each with probability $\frac{1}{6}$.

Thus,

$$E[X] = \sum_{x=1}^6 x \cdot P(X = x) = \left(1 \times \frac{1}{6}\right) + \left(2 \times \frac{1}{6}\right) + \left(3 \times \frac{1}{6}\right) + \left(4 \times \frac{1}{6}\right) + \left(5 \times \frac{1}{6}\right) + \left(6 \times \frac{1}{6}\right)$$
$$E[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

So, the expected value of rolling a die is **3.5**.

(B) Expected Value for a Continuous Random Variable

If X is a continuous random variable with probability density function (PDF) $f(x)$, the expected value is:

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx.$$

Example: Expected Value of an Exponential Distribution

For an exponential distribution with parameter λ , the PDF is:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

The expected value is:

$$E[X] = \int_0^{\infty} x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}.$$

If $\lambda = 0.2$ (e.g., waiting time between bus arrivals), then:

$$E[X] = \frac{1}{0.2} = 5.$$

This means the average waiting time is **5 minutes**.

2. Properties of Expectation

(A) Linearity of Expectation

For any two random variables X and Y , and constants a and b :

$$E[aX + bY] = aE[X] + bE[Y].$$

Example: Sum of Two Dice

Let X_1 and X_2 be the outcomes of two fair dice rolls. We know:

$$E[X_1] = 3.5, \quad E[X_2] = 3.5.$$

Thus, the expected sum:

$$E[X_1 + X_2] = E[X_1] + E[X_2] = 3.5 + 3.5 = 7.$$

(B) Expectation of a Constant

If c is a constant, then:

$$E[c] = c.$$

Example

If you always win ₹5 in a game, your expected winnings are:

$$E[X] = 5.$$

(C) Expectation of a Function of a Random Variable

If $g(X)$ is a function of X , then:

$$E[g(X)] = \sum g(x)P(X = x) \quad (\text{for discrete variables})$$

or

$$E[g(X)] = \int g(x)f(x)dx \quad (\text{for continuous variables}).$$

Example: Squaring the Outcome of a Die

Let X be a die roll. We compute $E[X^2]$:

$$E[X^2] = \sum_{x=1}^6 x^2 P(X = x).$$

$$E[X^2] = \left(1^2 \times \frac{1}{6}\right) + \left(2^2 \times \frac{1}{6}\right) + \left(3^2 \times \frac{1}{6}\right) + \left(4^2 \times \frac{1}{6}\right) + \left(5^2 \times \frac{1}{6}\right) + \left(6^2 \times \frac{1}{6}\right).$$

$$= \frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6} \approx 15.17.$$

3. Special Cases of Expectation

(A) Expectation of a Binomial Distribution

If $X \sim \text{Bin}(n, p)$, meaning X follows a **binomial distribution** with n trials and success probability p , then:

$$E[X] = np.$$

Example: Coin Tosses

If you flip a fair coin 10 times, the expected number of heads is:

$$E[X] = 10 \times \frac{1}{2} = 5.$$

(B) Expectation of a Poisson Distribution

If $X \sim \text{Poisson}(\lambda)$, then:

$$E[X] = \lambda.$$

Example: Calls to a Call Center

If a call center receives an average of 5 calls per hour, the expected number of calls in an hour is:

$$E[X] = 5.$$

4. Summary

Random Variable Type	Expectation Formula
Discrete X	$E[X] = \sum x P(X = x)$
Continuous X	$E[X] = \int x f(x) dx$
Binomial $X \sim \text{Bin}(n, p)$	$E[X] = np$
Poisson $X \sim \text{Poisson}(\lambda)$	$E[X] = \lambda$
Exponential $X \sim \text{Exp}(\lambda)$	$E[X] = \frac{1}{\lambda}$

Key Takeaways

- **Expected value** ($E[X]$) is the **average** outcome of a random variable over many trials.
- **For discrete variables**, sum over all possible values with their probabilities.
- **For continuous variables**, integrate over the probability density function.
- **Special distributions like Binomial, Poisson, and Exponential have predefined formulas** for expectation.

Joint and Marginal Probability Distributions

Probability distributions help us understand how variables behave. When dealing with multiple random variables, we use **joint probability distributions** and **marginal probability distributions** to describe their relationships.

1. Joint Probability Distribution

Definition

A **joint probability distribution** describes the probability of two (or more) random variables occurring together. It is used for **multiple dependent or independent random variables**.

For two discrete random variables X and Y , the joint probability mass function (PMF) is:

$$P(X = x, Y = y) = P(X, Y).$$

For continuous random variables, the joint probability density function (PDF) is:

$$f(X, Y) = \frac{\partial^2 P(X, Y)}{\partial x \partial y}.$$

Example (Discrete Case: Rolling Two Dice)

Let X and Y be the outcomes of rolling two fair six-sided dice. The joint probability distribution is:

$$P(X = 2, Y = 3) = P(X = 2) \cdot P(Y = 3) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}.$$

This means the probability of rolling a **2 on the first die** and a **3 on the second die** is $\frac{1}{36}$.

2. Marginal Probability Distribution

Definition

The **marginal probability distribution** is the probability of one variable **regardless of the other**. It is obtained by **summing (discrete case) or integrating (continuous case) the joint probability distribution** over the other variable.

For a discrete random variable X :

$$P(X = x) = \sum_y P(X = x, Y = y).$$

For a continuous random variable X :

$$f_X(x) = \int_{-\infty}^{\infty} f(X, Y) dy.$$

Example (Rolling Two Dice - Marginal Probability of X)

The marginal probability of rolling a particular value for **one die** (say X) is:

$$P(X = 2) = \sum_{y=1}^6 P(X = 2, Y = y) = \sum_{y=1}^6 \frac{1}{36} = \frac{6}{36} = \frac{1}{6}.$$

This shows that the probability of rolling **2 on the first die** (without considering the second die) is $\frac{1}{6}$.

3. Relationship Between Joint and Marginal Distributions

- **Marginal probability distributions** are derived from **joint probability distributions** by summing (discrete) or integrating (continuous) over the other variable.
- **If two variables are independent**, their joint distribution equals the product of their marginal distributions:

$$P(X, Y) = P(X)P(Y).$$

Example (Independent vs. Dependent Variables)

- **Independent Case (Fair Dice Roll):**

Since each die roll is independent,

$$P(X = 2, Y = 3) = P(X = 2)P(Y = 3).$$

- **Dependent Case (Weather & Umbrella Usage):**

If X is the event "rains" and Y is the event "uses umbrella," their joint probability is not just a product:

$$P(X, Y) \neq P(X)P(Y).$$

because people are more likely to use an umbrella when it rains.

4. Summary Table

Concept	Discrete Case	Continuous Case
Joint Probability $P(X, Y)$	$P(X = x, Y = y)$	$f(X, Y)$
Marginal Probability $P(X)$	$P(X) = \sum_y P(X, Y)$	$f_X(x) = \int f(X, Y) dy$
Independence	$P(X, Y) = P(X)P(Y)$	$f(X, Y) = f_X(x)f_Y(y)$

Properties of Expectation and Variance with Proofs

Expectation (also called **mathematical expectation or mean**) and **variance** are fundamental concepts in probability and statistics. Let's go through their important properties along with proofs.

1) Expectation (Mean) and Its Properties

The expectation $E[X]$ of a random variable X represents its average value over many trials.

Property 1: Linearity of Expectation

For any two random variables X and Y and constants a and b ,

$$E[aX + bY] = aE[X] + bE[Y].$$

Proof: Discrete Case

By the definition of expectation,

$$E[X] = \sum_x x P(X = x), \quad E[Y] = \sum_y y P(Y = y).$$

Now, for any constants a and b ,

$$E[aX + bY] = \sum_{x,y} (aX + bY)P(X = x, Y = y).$$

Using the sum property,

$$E[aX + bY] = \sum_{x,y} (aX)P(X = x, Y = y) + \sum_{x,y} (bY)P(X = x, Y = y).$$

Since constants can be factored out,

$$\begin{aligned} E[aX + bY] &= a \sum_{x,y} X P(X = x, Y = y) + b \sum_{x,y} Y P(X = x, Y = y). \\ &= aE[X] + bE[Y]. \end{aligned}$$

Corollary: Expectation of a constant is the constant itself, i.e.,

$$E[c] = c.$$

Property 2: Expectation of the Sum of Random Variables

For any n random variables $X_1, X_2, \dots, X_n$,

$$E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i].$$

Property 3: Expectation of a Product (If Independent)

If X and Y are independent,

$$E[XY] = E[X]E[Y].$$

Proof: Discrete Case

By definition,

$$E[XY] = \sum_{x,y} xy P(X = x, Y = y).$$

Since X and Y are independent, we use

$$P(X = x, Y = y) = P(X = x)P(Y = y).$$

Thus,

$$E[XY] = \sum_{x,y} xy P(X = x)P(Y = y).$$

Rearrange the sums:

$$E[XY] = \left(\sum_x x P(X = x) \right) \left(\sum_y y P(Y = y) \right) \\ = E[X]E[Y].$$

2) Variance and Its Properties

The variance $Var(X)$ of a random variable X measures its **spread** around the mean:

$$Var(X) = E[(X - E[X])^2].$$

Property 1: Alternative Formula for Variance

$$Var(X) = E[X^2] - (E[X])^2.$$

Proof

Expanding the definition of variance:

$$Var(X) = E[(X - E[X])^2].$$

Expanding the square:

$$= E[X^2 - 2XE[X] + (E[X])^2].$$

Using **linearity of expectation**:

$$Var(X) = E[X^2] - 2E[X]E[X] + (E[X])^2. \\ = E[X^2] - (E[X])^2.$$

Property 2: Variance of a Constant

$$Var(c) = 0, \quad \text{for any constant } c.$$

Proof

By definition,

$$Var(c) = E[(c - E[c])^2] = E[(c - c)^2] = E[0] = 0.$$

Property 3: Scaling Property of Variance

For a random variable X and a constant a :

$$Var(aX) = a^2 Var(X).$$

Proof

By definition of variance:

$$Var(aX) = E[(aX - E[aX])^2].$$

Using linearity of expectation:

$$E[aX] = aE[X].$$

Substituting:

$$\text{Var}(aX) = E[(aX - aE[X])^2] = E[a^2(X - E[X])^2].$$

Since a^2 is a constant:

$$\text{Var}(aX) = a^2 E[(X - E[X])^2] = a^2 \text{Var}(X).$$

Property 4: Variance of the Sum (If Independent)

For **independent** random variables X and Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Proof

Using the definition of variance:

$$\text{Var}(X + Y) = E[(X + Y - E[X + Y])^2].$$

By **linearity of expectation**:

$$E[X + Y] = E[X] + E[Y].$$

So,

$$\text{Var}(X + Y) = E[(X - E[X] + Y - E[Y])^2].$$

Expanding the square:

$$= E[(X - E[X])^2 + (Y - E[Y])^2 + 2(X - E[X])(Y - E[Y])].$$

Using **linearity of expectation**:

$$= E[(X - E[X])^2] + E[(Y - E[Y])^2] + 2E[(X - E[X])(Y - E[Y])].$$

If X and Y are **independent**, then:

$$E[(X - E[X])(Y - E[Y])] = 0.$$

Thus,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Final Summary Table

Property	Formula
Linearity of Expectation	$E[aX + bY] = aE[X] + bE[Y]$
Expectation of a Constant	$E[c] = c$
Expectation of a Product (Independent)	$E[XY] = E[X]E[Y]$
Alternative Variance Formula	$\text{Var}(X) = E[X^2] - (E[X])^2$
Variance of a Constant	$\text{Var}(c) = 0$
Scaling Property of Variance	$\text{Var}(aX) = a^2$
Variance of Sum (Independent Variables)	$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Theoretical Probability Distributions

A **probability distribution** describes how probabilities are assigned to different outcomes of a random variable. Theoretical probability distributions are **mathematically defined functions** that help model real-world situations.

Probability distributions are categorized into two types:

1. **Discrete Probability Distributions** (for discrete random variables)
2. **Continuous Probability Distributions** (for continuous random variables)

1. Discrete Probability Distributions

A **discrete probability distribution** applies to random variables that take **countable values** (e.g., 0, 1, 2, ...).

(a) Bernoulli Distribution

A random variable X follows a **Bernoulli distribution** if it has only **two possible outcomes**: Success (1) or Failure (0).

- **Probability Mass Function (PMF):**

$$P(X = x) = \begin{cases} p, & \text{if } x = 1 \\ 1 - p, & \text{if } x = 0 \end{cases}$$

where p is the probability of success.

- **Mean:** $E[X] = p$
- **Variance:** $Var(X) = p(1 - p)$

Example: Tossing a fair coin, where success is getting heads ($p = 0.5$) and failure is getting tails ($1 - p = 0.5$).

(b) Binomial Distribution

A **binomial distribution** models the number of successes in **n independent Bernoulli trials**.

- **PMF:**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$$

where

- n = number of trials
 - k = number of successes
 - p = probability of success in each trial
- **Mean:** $E[X] = np$
 - **Variance:** $Var(X) = np(1 - p)$

Example:

If a fair die is rolled 10 times, the probability of rolling a "4" in each roll is $p = \frac{1}{6}$. The number of times "4" appears follows a **Binomial distribution**.

(c) Poisson Distribution

A **Poisson distribution** models the number of events occurring in a fixed interval of time or space **when events happen at a constant rate** and are independent.

- **PMF:**

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

where

- λ = average number of occurrences per interval
- **Mean:** $E[X] = \lambda$
- **Variance:** $Var(X) = \lambda$

Example:

The number of customer arrivals at a store in an hour follows a Poisson distribution if arrivals are independent.

2. Continuous Probability Distributions

A **continuous probability distribution** applies to random variables that take **uncountable values** (e.g., height, weight, temperature).

(a) Uniform Distribution

A **uniform distribution** describes a situation where **all outcomes in an interval are equally likely**.

- **Probability Density Function (PDF):**

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

where a and b define the range of values.

- **Mean:** $E[X] = \frac{a+b}{2}$
- **Variance:** $Var(X) = \frac{(b-a)^2}{12}$

Example:

If a random number is chosen between 1 and 10, it follows a **Uniform distribution**.

(b) Normal (Gaussian) Distribution

A **normal distribution** models **many natural phenomena** (e.g., heights, test scores, errors in measurements).

- **PDF:**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where

- μ = mean
- σ^2 = variance
- **Mean:** $E[X] = \mu$
- **Variance:** $Var(X) = \sigma^2$

Example:

IQ scores follow a **Normal distribution** with a mean of 100 and standard deviation of 15.

(c) Exponential Distribution

An **exponential distribution** models the time between independent events that occur at a constant rate.

- **PDF:**

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- **Mean:** $E[X] = \frac{1}{\lambda}$
- **Variance:** $Var(X) = \frac{1}{\lambda^2}$

Example

The time between two consecutive calls received at a call center follows an **Exponential distribution**.

Summary Table of Theoretical Distributions

Distribution	Type	Parameters	PDF / PMF	Mean	Variance
Bernoulli	Discrete	p	$P(X = x) = p^x(1 - p)^{1-x}$	p	$p(1 - p)$
Binomial	Discrete	n, p	$P(X = k) = \binom{n}{k} p^k(1 - p)^{n-k}$	np	$np(1 - p)$
Poisson	Discrete	λ	$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ
Uniform	Continuous	a, b	$f(x) = \frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	Continuous	μ, σ^2	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Exponential	Continuous	λ	$f(x) = \lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Binomial Distribution

The **Binomial Distribution** is a **discrete probability distribution** that models the number of **successes** in a fixed number of **independent** trials, where each trial has only **two possible outcomes**: Success or Failure.

1. Definition of Binomial Distribution

A random variable X follows a **Binomial Distribution** if it counts the number of successes in n independent trials of an experiment, where:

- **Each trial has only two possible outcomes**: Success (S) or Failure (F).
- **The probability of success remains constant** across trials, denoted by p .
- **The trials are independent**, meaning the outcome of one trial does not affect another.

2. Probability Mass Function (PMF)

The probability of observing exactly k successes in n trials is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where:

- $\binom{n}{k} = \frac{n!}{k!(n-k)!} \rightarrow$ **Binomial coefficient** (counts the ways to choose k successes from n trials)
- $p \rightarrow$ Probability of success in a single trial
- $1 - p \rightarrow$ Probability of failure in a single trial
- $k \rightarrow$ Number of successes in n trials

Example Calculation:

Suppose we flip a fair coin **5 times** ($n = 5$), and we want to find the probability of getting **exactly 3 heads** ($k = 3$), where $p = 0.5$:

$$\begin{aligned} P(X = 3) &= \binom{5}{3} (0.5)^3 (0.5)^2 \\ &= \frac{5!}{3!(5-3)!} \times (0.5)^3 \times (0.5)^2 \\ &= \frac{5 \times 4}{2 \times 1} \times 0.125 \times 0.25 \\ &= 10 \times 0.03125 \\ &= 0.3125 \end{aligned}$$

Thus, the probability of getting **exactly 3 heads in 5 flips** is **0.3125** or **31.25%**.

3. Mean, Variance, and Standard Deviation

For a **Binomially distributed** random variable $X \sim B(n, p)$, the following formulas hold:

- **Mean (Expected Value):**

$$E[X] = np$$

- **Variance:**

$$Var(X) = np(1 - p)$$

- **Standard Deviation:**

$$\sigma = \sqrt{np(1 - p)}$$

Example:

A die is rolled **10 times** ($n = 10$), and we are interested in rolling a "6", so $p = \frac{1}{6}$.

- **Mean:**

$$E[X] = np = 10 \times \frac{1}{6} = \frac{10}{6} \approx 1.67$$

This means we expect around **1.67 sixes** in 10 rolls.

- **Variance:**

$$Var(X) = np(1 - p) = 10 \times \frac{1}{6} \times \frac{5}{6} = 1.39$$

- **Standard Deviation:**

$$\sigma = \sqrt{1.39} \approx 1.18$$

This means that the number of times a "6" appears will typically be **within 1.18 of the mean (1.67)**.

4. Real-Life Applications of Binomial Distribution

(a) Quality Control in Manufacturing

A factory produces **LED bulbs**, and **5% of bulbs** are defective. If we randomly pick **20 bulbs**, what is the probability that **exactly 2 bulbs are defective**?

- $n = 20$ (total bulbs)
- $p = 0.05$ (probability of defective bulb)
- $k = 2$ (we want exactly 2 defective bulbs)

Using the **Binomial PMF formula**:

$$\begin{aligned} P(X = 2) &= \binom{20}{2} (0.05)^2 (0.95)^{18} \\ &= \frac{20!}{2! (18!)} (0.0025) (0.377) \\ &= 190 \times 0.0009425 = 0.179 \end{aligned}$$

So, the probability of **exactly 2 defective bulbs** is **17.9%**.

(b) Medical Trials

A new vaccine is **80% effective**. If 10 people receive the vaccine, what is the probability that **exactly 8 people** are protected?

- $n = 10$ (total vaccinated people)
- $p = 0.88$ (probability of protection)
- $k = 8$ (want exactly 8 protected)

Using **Binomial PMF**:

$$\begin{aligned} P(X = 8) &= \binom{10}{8} (0.8)^8 (0.2)^2 \\ &= 45 \times 0.1678 \times 0.04 \\ &= 0.302 \end{aligned}$$

So, the probability of **exactly 8 people being protected** is **30.2%**.

5. Properties of Binomial Distribution

1. **If $p = 0.5$, the binomial distribution is symmetric.**
 - Example: Tossing a fair coin multiple times.
2. **If $p > 0.5$, the distribution is skewed to the left (more successes).**
 - Example: A biased coin with heads more likely than tails.
3. **If $p < 0.5$, the distribution is skewed to the right (fewer successes).**
 - Example: A basketball player making 30% of free throws.
4. **As n increases, the binomial distribution approximates a normal distribution (Central Limit Theorem).**
 - For large n , we can use: $X \sim N(np, np(1 - p))$

6. Conclusion

- The **Binomial distribution** models the number of successes in **independent** trials.
- It is defined by **two parameters**: n (number of trials) and p (success probability).
- The probability of exactly k successes is computed using the **PMF formula**.
- The mean is np variance is $np(1 - p)$, and standard deviation is $\sqrt{np(1 - p)}$.
- It has applications in **quality control, medical trials, genetics, marketing, and sports**.

Poisson Distribution

The **Poisson Distribution** is a **discrete probability distribution** that models the number of events occurring in a fixed interval of **time, space, or any measurable unit**, given that these events happen at a constant mean rate and independently of the time since the last event.

1. Definition of Poisson Distribution

A random variable X follows a **Poisson Distribution** if it represents the number of events occurring in a fixed interval, where:

- Events occur **independently** (one event does not affect another).
- The probability of more than one event occurring in a very small interval is **negligible**.
- The average rate of occurrence (events per unit time or space) is **constant**.

The **Poisson Distribution** is defined by a single parameter:

- $\lambda \rightarrow$ The **average rate** (mean) of events occurring per interval.

2. Probability Mass Function (PMF)

The probability of exactly k events occurring in a fixed interval is given by:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, 3, \dots$$

where:

- $e \approx 2.718 \rightarrow$ Euler's number (mathematical constant)
- $\lambda \rightarrow$ The **expected number** of occurrences in the interval
- $k! \rightarrow$ Factorial of k , representing the number of ways k events can occur.

Example Calculation:

A call center receives an average of **3 calls per hour** ($3\lambda = 3$). What is the probability that exactly **5 calls** arrive in an hour ($k = 5$)?

$$\begin{aligned} P(X = 5) &= \frac{e^{-3} 3^5}{5!} \\ &= \frac{0.0498 \times 243}{120} \\ &= \frac{12.095}{120} \\ &= 0.1008 \end{aligned}$$

Thus, the probability of receiving **exactly 5 calls** in an hour is **10.08%**.

3. Mean, Variance, and Standard Deviation

For a Poisson-distributed random variable $X \sim P(\lambda)$:

- **Mean (Expected Value):**

$$E[X] = \lambda$$

- **Variance:**

$$Var(X) = \lambda$$

- **Standard Deviation:**

$$\sigma = \sqrt{\lambda}$$

Example:

If a hospital receives **8 emergency cases per day** on average ($8\lambda=8$):

- **Mean:** $E[X] = 8$
- **Variance:** $Var(X) = 8$
- **Standard Deviation:** $\sigma = \sqrt{8} \approx 2.83$

This means that the **actual number of emergency cases per day** will typically vary by about **2.83 cases from the mean (8 cases)**.

4. Real-Life Applications of Poisson Distribution

(a) Customer Arrival at a Bank

A bank receives an average of **10 customers per hour**. What is the probability that **exactly 7 customers** arrive in the next hour?

- $\lambda = 10$ (average number of arrivals)
- $k = 7$ (exact number of customers)

Using the **Poisson PMF formula**:

$$\begin{aligned} P(X = 7) &= \frac{e^{-10} 10^7}{7!} \\ &= \frac{0.0000454 \times 10,000,000}{5040} \\ &= 0.0902 \end{aligned}$$

So, the probability of exactly **7 customers arriving in an hour** is **9.02%**.

(b) Network Packet Transmission

A router receives **15 data packets per second** on average. What is the probability that it receives **exactly 20 packets in a second**?

- $\lambda = 15$
- $k = 20$

$$\begin{aligned} P(X = 20) &= \frac{e^{-15} 15^{20}}{20!} \\ &= \frac{0.0000302 \times 3.2768 \times 10^{23}}{2.4329 \times 10^{18}} = 0.0516 \end{aligned}$$

Thus, the probability of receiving **exactly 20 packets** in a second is **5.16%**.

5. Poisson Approximation to the Binomial Distribution

When **n is large** and **p is small** in a **Binomial distribution**, we can approximate it using a Poisson distribution:

$$X \sim \text{Bin}(n, p) \text{ is approximated by } P(\lambda) \text{ where } \lambda = np$$

Example:

A factory produces **1000 bulbs** daily, and each bulb has a **0.002 probability of being defective**.

- **Binomial parameters:** $n = 1000, p = 0.002$
- **Poisson approximation:** $\lambda = np = 1000 \times 0.002 = 2$

Thus, we can approximate this binomial distribution with $P(2)$

6. Properties of Poisson Distribution

1. The mean and variance are equal:

$$E[X] = Var(X) = \lambda$$

2. If two independent Poisson distributions $P(\lambda_1)$ and $P(\lambda_2)$ are combined, the resulting distribution is also Poisson:

$$X_1 + X_2 \sim P(\lambda_1 + \lambda_2)$$

Example:

If a hospital receives **5 emergency cases per hour** in the morning ($\lambda_1 = 5$) and **7 per hour** in the afternoon ($\lambda_2 = 7$), the total arrivals follow:

$$X \sim P(5 + 7) = P(12)$$

3. Poisson distribution is a limiting case of Binomial distribution:

- When $n \rightarrow \infty$ and $p \rightarrow 0$, but np remains constant, the binomial distribution approaches a Poisson distribution.

7. Conclusion

- The **Poisson Distribution** models the number of events occurring in a **fixed interval** when events happen at a **constant mean rate**.
- It is **defined by one parameter** λ , which represents the **expected number of occurrences**.
- The probability of exactly k events is computed using the **PMF formula**.
- The mean and variance are both equal to λ .
- It is widely used in **real-world applications** like **call centers, hospital cases, customer arrivals, traffic flow, and network packet transmission**.
- It can approximate the **Binomial distribution** when nnn is large and ppp is small.

Normal Distribution

The **Normal Distribution**, also known as the **Gaussian Distribution**, is a **continuous probability distribution** that is widely used in statistics and machine learning. It describes data that clusters around a central value with no bias to the left or right, forming a **bell-shaped curve**.

1. Definition of Normal Distribution

A **random variable** X follows a **Normal Distribution** if its probability density function (PDF) is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where:

- $\mu \rightarrow$ **Mean** (center of the distribution)
- $\sigma \rightarrow$ **Standard deviation** (spread of the distribution)
- $\sigma^2 \rightarrow$ **Variance** (square of the standard deviation)
- $e \approx 2.718$ (Euler's number)

- $\pi \approx 3.1416$

Key Characteristics:

- The curve is **bell-shaped** and **symmetric** around μ .
- The total area under the curve is **1**.
- The probability of a value being far from μ decreases exponentially.

2. Properties of Normal Distribution

1. **Symmetry:** The curve is symmetric around the mean μ , meaning $P(X > \mu) = P(X < \mu)$.
2. **Empirical Rule (68-95-99.7 Rule):**
 - **68%** of the data falls within **1 standard deviation** ($\mu \pm \sigma$).
 - **95%** of the data falls within **2 standard deviations** ($\mu \pm 2\sigma$).
 - **99.7%** of the data falls within **3 standard deviations** ($\mu \pm 3\sigma$).
3. **Mean, Median, and Mode are the Same:**
 - Since it is symmetric, the mean μ , median, and mode are equal.
4. **Asymptotic Nature:**
 - The curve approaches the x-axis but never touches it.

3. Standard Normal Distribution (Z-Score Transformation)

A **Standard Normal Distribution** is a normal distribution with:

- Mean $\mu = 0$
- Standard deviation $\sigma = 1$

To convert any normal distribution $X \sim N(\mu, \sigma^2)$ to a **Standard Normal Distribution**, we use the **Z-score formula**:

$$Z = \frac{X - \mu}{\sigma}$$

where:

- X is the original value.
- Z is the number of standard deviations X is away from μ .

Example: If a student scores **80 marks** in an exam where the class mean is **70** and the standard deviation is **10**, the **Z-score** is:

$$Z = \frac{80 - 70}{10} = \frac{10}{10} = 1$$

This means the student's score is **1 standard deviation above the mean**.

4. Real-Life Examples of Normal Distribution

(a) Heights of People

Human heights in a large population follow a **normal distribution**.

- Suppose **adult male heights** follow $N(175, 10^2)$, meaning the average height is **175 cm** with a **standard deviation of 10 cm**.
- The probability of a randomly chosen male being **between 165 cm and 185 cm** can be found using the **Z-score**.

$$Z = \frac{165 - 175}{10} = -1, \quad Z = \frac{185 - 175}{10} = 1$$

From Z-tables, the probability of **Z between -1 and 1** is **68%**.

Thus, **68% of men are between 165 cm and 185 cm**.

(b) Exam Scores

A university entrance exam has scores following $N(500, 100^2)$.

- If **top 5%** of students receive a scholarship, what is the minimum required score?

Using a **Z-table**, the top 5% corresponds to $Z = 1.645$.

Using the **Z-score formula**:

$$X = \mu + Z\sigma = 500 + (1.645 \times 100) = 664.5$$

Thus, a student needs at least **665 marks** to be in the top 5%.

(c) Measurement Errors

- Measurement errors in scientific experiments often follow a **normal distribution**, with most errors clustering around the true value.

5. Central Limit Theorem (CLT) and Normal Distribution

The **Central Limit Theorem (CLT)** states:

"The distribution of the sample mean approaches a **normal distribution**, regardless of the original population distribution, as the sample size increases."

This is why **many real-world processes** follow a **normal distribution**.

Example:

- If we measure the **average weight** of **10,000 people**, the **distribution of sample means** will be approximately **normal**, even if individual weights are **not normally distributed**.

6. Conclusion

- **Normal Distribution** is a **continuous probability distribution** with a **bell-shaped curve**.
- It is **symmetrical**, with **mean = median = mode**.
- It follows the **68-95-99.7 rule** for standard deviations.
- Many **real-world phenomena** follow a **normal distribution**: heights, exam scores, errors, and financial data.
- The **Z-score transformation** helps convert any normal distribution to a **standard normal distribution**.
- The **Central Limit Theorem (CLT)** explains why normal distribution appears frequently in statistics.

Fitting of Binomial Distribution: Explanation with Example

1. Introduction to Fitting a Binomial Distribution

Fitting a **binomial distribution** means estimating the parameters of a **binomial model** from given observed data. This involves:

1. **Determining whether the given data follows a binomial distribution.**
2. **Estimating the parameters:**
 - $n \rightarrow$ **Number of trials**
 - $p \rightarrow$ **Probability of success**
3. **Comparing observed and expected frequencies** to check the goodness of fit.

2. Conditions for Binomial Distribution

A binomial distribution applies if:

- There are **fixed** n independent trials.
- Each trial has **only two possible outcomes**: success (S) or failure (F).
- The probability of success (p) remains **constant** for all trials.
- The trials are **independent**.

3. Steps to Fit a Binomial Distribution

Given a dataset with observed frequencies of successes, follow these steps:

Step 1: Calculate Mean ($\bar{x}$)

$$\bar{x} = \frac{\sum(x \cdot f)}{N}$$

where:

- $x \rightarrow$ Number of successes
- $f \rightarrow$ Frequency of occurrences
- $N \rightarrow$ Total number of trials ($N = \sum f$)

Step 2: Estimate Probability of Success (p)

Since the **binomial mean** is given by:

$$\mu = np$$

We estimate p as:

$$p = \frac{\bar{x}}{n}$$

where n is the **maximum number of successes observed**.

Step 3: Compute Binomial Probabilities

The probability of exactly x successes is given by the **Binomial Probability Formula**:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

where:

- $\binom{n}{x} = \frac{n!}{x!(n-x)!}$ is the **binomial coefficient**.
- p is the probability of success.

Step 4: Compute Expected Frequencies

Expected frequency for each x :

$$E(x) = P(X = x) \times N$$

where N is the total number of trials.

Step 5: Compare Observed and Expected Frequencies

To check if the data fits the binomial distribution, we compare:

- **Observed Frequencies (O)**
- **Expected Frequencies (E)**

A **Chi-square goodness-of-fit test** can be used for further validation:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

If χ^2 is small, the binomial model is a good fit.

4. Example: Fitting a Binomial Distribution

Given Data:

A fair coin is tossed **5 times per trial**, and the number of heads observed is recorded. The following data is collected from **100 trials**.

x (Heads) 0 1 2 3 4 5 Total

Frequency (f) 5 15 30 25 20 5 **100**

Step 1: Calculate Mean ($\bar{x}$)

$$\begin{aligned} \bar{x} &= \frac{\sum(x \cdot f)}{N} = \frac{(0 \times 5) + (1 \times 15) + (2 \times 30) + (3 \times 25) + (4 \times 20) + (5 \times 5)}{100} \\ &= \frac{0 + 15 + 60 + 75 + 80 + 25}{100} = \frac{255}{100} = 2.55 \end{aligned}$$

Step 2: Estimate p

$$p = \frac{\bar{x}}{n} = \frac{2.55}{5} = 0.51$$

So, estimated probability of heads in each trial is $p = 0.51$.

Step 3: Compute Binomial Probabilities

The binomial probability formula is:

$$P(X = x) = \binom{5}{x} (0.51)^x (0.49)^{5-x}$$

For each x :

$$P(0) = \binom{5}{0} (0.51)^0 (0.49)^5 = 1 \times 1 \times 0.030 = 0.030$$

$$P(1) = \binom{5}{1} (0.51)^1 (0.49)^4 = 5 \times 0.51 \times 0.075 = 0.191$$

$$P(2) = \binom{5}{2} (0.51)^2 (0.49)^3 = 10 \times 0.2601 \times 0.117 = 0.260$$

$$P(3) = \binom{5}{3} (0.51)^3 (0.49)^2 = 10 \times 0.132 \times 0.240 = 0.252$$

$$P(4) = \binom{5}{4} (0.51)^4 (0.49)^1 = 5 \times 0.067 \times 0.49 = 0.164$$

$$P(5) = \binom{5}{5} (0.51)^5 (0.49)^0 = 1 \times 0.034 \times 1 = 0.034$$

Step 4: Compute Expected Frequencies

$$E(x) = P(X = x) \times 100$$

x	Observed f	Probability $P(x)$	Expected $E(x)$
0	5	0.030	3.0
1	15	0.191	19.1
2	30	0.260	26.0
3	25	0.252	25.2
4	20	0.164	16.4
5	5	0.034	3.4

Step 5: Compare Observed vs. Expected Frequencies

The observed frequencies are **close** to the expected values, meaning the **binomial distribution is a good fit**.

5. Conclusion

- **Fitting a binomial distribution** involves estimating **nnn** and **ppp**, then computing expected frequencies.
- If observed and expected frequencies **match closely**, the binomial model is a **good fit**.
- **Chi-square goodness-of-fit test** can further confirm the accuracy of the model.
- Binomial distribution is widely used in **quality control, genetics, and sports analytics**.

Properties of Binomial Distribution

The **binomial distribution** is a **discrete probability distribution** that models the number of **successes** in a **fixed number of independent trials**, where each trial has **only two possible outcomes** (success or failure).

1. Properties of Binomial Distribution

A **binomially distributed random variable** X with parameters nnn (number of trials) and ppp (probability of success) is denoted as:

$$X \sim B(n, p)$$

This means X follows a binomial distribution with **parameters nnn and p**.

Key Properties:

1. **The mean (expected value):** $E(X) = np$
2. **The variance:** $Var(X) = np(1 - p)$
3. **The standard deviation:** $\sigma = \sqrt{np(1 - p)}$
4. **The moment-generating function (MGF):** $M_X(t) = (1 - p + pe^t)^n$
5. **The skewness:** $\frac{1-2p}{\sqrt{np(1-p)}}$
6. **The kurtosis:** $\frac{1-6p(1-p)}{np(1-p)}$
7. **The probability mass function (PMF):** $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n$
8. **The cumulative distribution function (CDF):** $F(k) = P(X \leq k) = \sum_{x=0}^k P(X = x)$
9. **If $p = 0.5$, the distribution is symmetric; otherwise, it is skewed.**

2. Proofs of Important Properties

Property 1: Mean (Expected Value)

$$E(X) = np$$

Proof:

The expected value (mean) of X is given by:

$$E(X) = \sum_{k=0}^n k P(X = k)$$

Using properties of binomial coefficients and summation, we arrive at:

$$E(X) = np$$

Example:

Suppose a coin is flipped **10 times** ($n = 10$) with a probability of heads $p = 0.5$. The expected number of heads is:

$$E(X) = np = 10 \times 0.5 = 5$$

Property 2: Variance

$$Var(X) = np(1 - p)$$

Proof:

Variance is given by:

$$Var(X) = E(X^2) - (E(X))^2$$

After simplifications using the binomial expansion, we obtain:

$$Var(X) = np(1 - p)$$

Example:

For $n = 10, p = 0.5$,

$$Var(X) = 10 \times 0.5 \times 0.5 = 2.5$$

Property 3: Standard Deviation

$$\sigma = \sqrt{Var(X)} = \sqrt{np(1-p)}$$

Example:

For $n = 10, p = 0.5$,

$$\sigma = \sqrt{10 \times 0.5 \times 0.5} = \sqrt{2.5} \approx 1.58$$

Property 4: Skewness

$$\text{Skewness} = \frac{1 - 2p}{\sqrt{np(1-p)}}$$

- If $p = 0.5$, the binomial distribution is **symmetrical**.
- If $p < 0.5$, it is **right-skewed**.
- If $p > 0.5$, it is **left-skewed**.

Example:

For $n = 10, p = 0.5$,

$$\text{Skewness} = \frac{1 - 2(0.5)}{\sqrt{10(0.5)(0.5)}} = 0$$

Since the skewness is **0**, the binomial distribution is **symmetric**.

Property 5: Kurtosis

$$\text{Kurtosis} = \frac{1 - 6p(1-p)}{np(1-p)}$$

Kurtosis measures the "**tailedness**" of the distribution.

- If **kurtosis** > 3 , the distribution has heavy tails (**leptokurtic**).
- If **kurtosis** < 3 , the distribution has light tails (**platykurtic**).
- If **kurtosis** $= 3$, the distribution is **mesokurtic**.

Example:

For $n = 10, p = 0.5$,

$$\text{Kurtosis} = \frac{1 - 6(0.5)(0.5)}{10(0.5)(0.5)} = 0.4$$

Since $0.4 < 3$, the binomial distribution is **platykurtic**.

3. Example: Properties in Action

Example Scenario

A factory produces **light bulbs**, and each bulb has a **5% defect rate**. A random sample of **20 bulbs** is tested.

1. What is the probability of exactly 3 defective bulbs?

- $n = 20, p = 0.05$
- Using the PMF formula: $P(X = 3) = \binom{20}{3}(0.05)^3(0.95)^{17}$
- $= \frac{20!}{3!(17!)}(0.05)^3(0.95)^{17}$
- ≈ 0.185

2. What is the expected number of defective bulbs?

$$E(X) = np = 20 \times 0.05 = 1$$

3. What is the variance?

$$Var(X) = np(1 - p) = 20 \times 0.05 \times 0.95 = 0.95$$

4. What is the standard deviation?

$$\sigma = \sqrt{Var(X)} = \sqrt{0.95} \approx 0.975$$

4. Summary Table

Property	Formula	Example Calculation ($n = 10, p = 0.5$)
Mean	np	$10 \times 0.5 = 5$
Variance	$np(1 - p)$	$10 \times 0.5 \times 0.5 = 2.5$
Standard Deviation	$\sqrt{np(1 - p)}$	$\sqrt{2.5} \approx 1.58$
Skewness	$\frac{1 - 2p}{\sqrt{np(1 - p)}}$	0 (Symmetric)
Kurtosis	$\frac{1 - 6p(1 - p)}{np(1 - p)}$	0.4 (Platykurtic)

Poisson and Normal Distributions

1. Poisson Distribution

The **Poisson distribution** is a **discrete probability distribution** that models the number of times an event occurs in a fixed interval of time or space when:

1. The events occur **independently**.
2. The average number of occurrences (λ) in the given interval is known.
3. The probability of more than one event occurring in a very small interval is negligible.

Probability Mass Function (PMF)

The probability of exactly **k** occurrences is given by:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, 3, \dots$$

where:

- λ (lambda) is the expected number of occurrences in the given interval.

- e is Euler's number (≈ 2.718).
- $k!$ is the factorial of k .

Properties of Poisson Distribution

1. **Mean (Expected Value):** $E(X) = \lambda$
2. **Variance:** $Var(X) = \lambda$
3. **Standard Deviation:** $\sigma = \sqrt{\lambda}$
4. **Skewness:** $\frac{1}{\sqrt{\lambda}}$ (right – skewed if λ is small)
5. **Poisson Approximation to Binomial:** If n is large and p is small, then $B(n, p) \approx \text{Poisson}(\lambda = np)$.

Example of Poisson Distribution

Scenario:

A call center receives **5 customer calls per hour** on average. What is the probability of receiving **exactly 3 calls** in an hour?

Solution:

Given:

- $\lambda = 5$ (expected number of calls per hour)
- $k = 3$ (number of calls we want to find probability for)

Using the Poisson formula:

$$\begin{aligned}
 P(X = 3) &= \frac{e^{-5} 5^3}{3!} \\
 &= \frac{e^{-5} \times 125}{6} \\
 &= \frac{0.0067 \times 125}{6} \\
 &= 0.1404 \\
 &= 0.1404
 \end{aligned}$$

Thus, the probability of receiving exactly **3 calls** in an hour is **0.1404 (or 14.04%)**.

2. Normal Distribution

The **normal distribution** is a **continuous probability distribution** that is symmetric around the mean. It is also known as the **Gaussian distribution**.

Probability Density Function (PDF)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where:

- μ = Mean (expected value)
- σ^2 = Variance
- σ = Standard deviation

- e = Euler's number (≈ 2.718)
- $\pi = Pi$ (≈ 3.1416)

Properties of Normal Distribution

1. **Mean = Median = Mode (μ).**
2. **Symmetrical about the mean.**
3. **Total area under the curve = 1.**
4. **68-95-99.7 Rule (Empirical Rule):**
 - **68%** of data lies within $\mu \pm \sigma$.
 - **95%** of data lies within $\mu \pm 2\sigma$.
 - **99.7%** of data lies within $\mu \pm 3\sigma$.
5. The curve approaches, but never touches, the x-axis.

Standard Normal Distribution

The **standard normal distribution** has:

- Mean $\mu = 0$
- Standard deviation $\sigma = 1$

It is used to compute probabilities using the **Z-score**, which is given by:

$$Z = \frac{X - \mu}{\sigma}$$

where:

- X = Raw score
- μ = Mean
- σ = Standard deviation

Example of Normal Distribution

Scenario:

The heights of students in a school follow a normal distribution with:

- Mean height: $\mu = 170 \text{ cm}$
- Standard deviation: $\sigma = 10 \text{ cm}$

Find the probability that a randomly selected student is **shorter than 160 cm**.

Solution:

Step 1: Compute the Z-score

$$Z = \frac{X - \mu}{\sigma} = \frac{160 - 170}{10} = \frac{-10}{10} = -1$$

Step 2: Use a standard normal table or calculator

From the Z-table, the probability corresponding to $Z = -1$ is **0.1587**.

Thus, the probability that a student is shorter than **160 cm** is **15.87%**.

3. Comparison: Poisson vs. Normal Distribution

Feature	Poisson Distribution	Normal Distribution
Type	Discrete	Continuous
Shape	Right-skewed for small λ , becomes symmetric for large λ	Symmetric (bell-shaped)
Mean	λ	μ
Variance	λ	σ^2
Application	Count of rare events (e.g., calls, accidents)	Natural phenomena (e.g., height, test scores)

4. Summary

1. Poisson Distribution:

- Models **counts** of events over time/space.
- Uses λ as the mean and variance.
- Example:** Number of calls per hour in a call center.

2. Normal Distribution:

- Models **continuous** data.
- Defined by μ (**mean**) and σ^2 (**variance**).
- Example:** Heights of students.

Relation Between Binomial and Normal Distributions

1. Introduction

The **binomial distribution** and the **normal distribution** are related in probability theory. Specifically, as the number of trials in a binomial experiment increases, the binomial distribution starts resembling the normal distribution. This relationship is essential because, in many real-world problems, we approximate a binomial distribution using a normal distribution when certain conditions are met.

2. Understanding Binomial Distribution

The **binomial distribution** is a **discrete probability distribution** that models the number of **successes** in n independent trials, each having a **success probability** p .

Probability Mass Function (PMF) of Binomial Distribution

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where:

- n = number of trials
- k = number of successes
- p = probability of success in each trial
- $(1 - p)$ = probability of failure

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ (binomial coefficient)

Mean and Variance of Binomial Distribution

- **Mean (Expected Value):** $E(X) = np$
- **Variance:** $Var(X) = np(1 - p)$

3. Understanding Normal Distribution

The **normal distribution** is a **continuous probability distribution** that is symmetric and bell-shaped. It is described by the probability density function (PDF):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where:

- μ = Mean
- σ^2 = Variance
- σ = Standard deviation

Standard Normal Distribution

When $\mu = 0$ and $\sigma = 1$, we get the **standard normal distribution**, and we use the **Z-score** to standardize values:

$$Z = \frac{X - \mu}{\sigma}$$

4. Relationship Between Binomial and Normal Distributions

When the number of trials n in a **binomial distribution** is large, the binomial distribution becomes approximately **normal**.

Normal Approximation to Binomial

For large n , the binomial distribution can be approximated by a normal distribution with:

- **Mean:** $\mu = np$
- **Standard deviation:** $\sigma = \sqrt{np(1 - p)}$

Conditions for Normal Approximation to Binomial (Central Limit Theorem)

The normal approximation works well when:

$$np \geq 5 \quad \text{and} \quad n(1 - p) \geq 5$$

This ensures that the binomial distribution is not too skewed.

5. Example of Binomial to Normal Approximation

Scenario

A factory produces light bulbs, and **each bulb has a 5% defect rate**. A **random sample of 200 bulbs** is taken. What is the probability that **more than 15 bulbs are defective**?

Step 1: Define Binomial Distribution Parameters

- $n = 200$ (number of bulbs)
- $p = 0.05$ (probability of a defective bulb)

- X = number of defective bulbs follows $Binomial(200, 0.05)$

Step 2: Compute Mean and Standard Deviation

$$\mu = np = 200 \times 0.05 = 10$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{200 \times 0.05 \times 0.95} = \sqrt{9.5} \approx 3.08$$

Step 3: Convert to Normal Distribution Using Z-score

We approximate X using $N(10, 3.08)$.

We need $P(X > 15)$, which we approximate as $P\left(Z > \frac{15.5-10}{3.08}\right)$ (using **continuity correction** by adding 0.5).

$$Z = \frac{15.5 - 10}{3.08} = \frac{5.5}{3.08} \approx 1.79$$

Step 4: Find Probability from Z-Table

From the standard normal table:

$$P(Z > 1.79) = 1 - P(Z < 1.79) = 1 - 0.9633 = 0.0367$$

Thus, the probability that **more than 15 bulbs are defective** is **3.67%**.

6. 6. Key Takeaways

1. Binomial and Normal distributions are related:

- The binomial distribution is **discrete**, while the normal distribution is **continuous**.
- As n increases, the binomial histogram **looks like a normal curve**.

2. Normal Approximation is useful:

- When n is large and p is not too close to 0 or 1, we can use a normal approximation to **simplify calculations**.

3. Continuity Correction is necessary:

- Since the binomial is discrete and normal is continuous, we adjust by **± 0.5** before using the normal approximation.

7. 7. Summary Table

Feature	Binomial Distribution	Normal Distribution
Type	Discrete	Continuous
Formula	$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
Mean	$\mu = np$	μ
Variance	$\sigma^2 = np(1-p)$	σ^2
When to Use	Counting successes in trials	Modeling continuous data
Approximation Condition	$np \geq 5, n(1-p) \geq 5$	Always valid

Relation Between Poisson and Normal Distributions

1. Introduction

The **Poisson distribution** and the **normal distribution** are related in probability theory. Specifically, when the mean of a Poisson distribution is **large**, the Poisson distribution starts resembling the **normal distribution**. This relationship is useful because it allows us to approximate a **Poisson-distributed** variable using the **normal distribution**, which is easier to work with in calculations and statistical applications.

2. Understanding Poisson Distribution

The **Poisson distribution** is a **discrete probability distribution** that models the number of events occurring in a **fixed interval of time or space**, assuming that:

- Events occur independently.
- The average rate of occurrence (λ) is constant.

Probability Mass Function (PMF) of Poisson Distribution

The Poisson probability of exactly k occurrences is:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

where:

- k = number of occurrences (0, 1, 2, ...)
- λ = expected number of occurrences in the given time/space
- e = Euler's number (≈ 2.718)

Mean and Variance of Poisson Distribution

- **Mean (Expected Value):** $E(X) = \lambda$
- **Variance:** $Var(X) = \lambda$

Thus, in a Poisson distribution, the **mean and variance are equal**.

3. Understanding Normal Distribution

The **normal distribution** is a **continuous probability distribution** that is symmetric and bell-shaped. It is described by the probability density function (PDF):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where:

- μ = Mean
- σ^2 = Variance
- σ = Standard deviation

4. Relationship Between Poisson and Normal Distributions

When the **mean (λ) is large** in a Poisson distribution, the shape of the distribution starts resembling a **normal distribution**.

Normal Approximation to Poisson

For large λ , the Poisson distribution can be approximated by a normal distribution with:

- **Mean:** $\mu = \lambda$
- **Standard deviation:** $\sigma = \sqrt{\lambda}$

Thus, the Poisson distribution $P(X = k)$ can be approximated using the normal distribution:

$$X \sim N(\lambda, \sqrt{\lambda})$$

Conditions for Normal Approximation to Poisson

The normal approximation works well when:

$$\lambda \geq 30$$

For smaller values of λ , the Poisson distribution remains **skewed** and does not resemble a normal distribution.

5. Example: Normal Approximation of Poisson

Scenario

Suppose a call center receives an **average of 50 calls per hour** ($\lambda = 50$). What is the probability that in the next hour, the center will receive **more than 60 calls**?

Step 1: Define Poisson Distribution Parameters

- $\lambda = 50$ (mean number of calls per hour)
- $X \sim \text{Poisson}(50)$

Step 2: Compute Mean and Standard Deviation

Since the Poisson distribution can be approximated by a normal distribution when λ is large:

- **Mean:** $\mu = \lambda = 50$
- **Standard deviation:** $\sigma = \sqrt{\lambda} = \sqrt{50} \approx 7.07$

Step 3: Convert to Normal Distribution Using Z-score

To find $P(X > 60)$, we use the normal approximation:

$$P(X > 60) = P\left(Z > \frac{60.5 - 50}{7.07}\right)$$

(Note: We use **continuity correction** by adding 0.5 to 60.)

$$Z = \frac{60.5 - 50}{7.07} = \frac{10.5}{7.07} \approx 1.49$$

Step 4: Find Probability from Z-Table

From the standard normal table:

$$P(Z > 1.49) = 1 - P(Z < 1.49) = 1 - 0.9319 = 0.0681$$

Thus, the probability that **more than 60 calls** are received in the next hour is **6.81%**.

6. Key Takeaways

1. **Poisson and Normal distributions are related:**
 - The **Poisson distribution is discrete**, while the **normal distribution is continuous**.

- For **large λ** , the Poisson distribution behaves **approximately like a normal distribution**.
2. **Normal Approximation is useful:**
- When $\lambda \geq 30$, the Poisson distribution is **symmetric** and can be **approximated using a normal distribution**.
 - This makes calculations easier, especially when working with probabilities.
3. **Continuity Correction is necessary:**
- Since the Poisson distribution is **discrete**, and the normal distribution is **continuous**, we adjust by ± 0.5 before using the normal approximation.

7. Summary Table

Feature	Poisson Distribution	Normal Distribution
Type	Discrete	Continuous
Formula	$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
Mean	$\mu = \lambda$	μ
Variance	$\sigma^2 = \lambda$	σ^2
When to Use	Modeling rare events	Modeling continuous data
Approximation Condition	$\lambda \geq 30$	Always valid

Importance of Normal Distribution

1. Introduction

The **normal distribution**, also known as the **Gaussian distribution**, is one of the most important probability distributions in statistics. It is widely used in real-world applications due to its **symmetry**, **bell-shaped curve**, and the **Central Limit Theorem (CLT)**, which states that the sum of many independent random variables tends to follow a normal distribution.

2. Definition of Normal Distribution

A random variable X follows a normal distribution if its probability density function (PDF) is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where:

- μ = **Mean (center of the distribution)**
- σ^2 = **Variance (spread of data)**
- σ = **Standard deviation**
- $e \approx 2.718$ = **Euler's number**

- $\pi \approx 3.1416$

3. Importance of Normal Distribution

1. Follows Central Limit Theorem (CLT)

- The CLT states that the sum (or average) of a large number of independent random variables follows a normal distribution, regardless of the original distribution of the data.
- This makes the normal distribution fundamental in inferential statistics.
- ♦ **Example:** If we repeatedly sample 100 students' exam scores from different schools, the mean scores of these samples will follow a normal distribution even if individual scores do not.

2. Common in Natural Phenomena

Many real-world measurements follow a normal distribution due to the effect of multiple small influences.

Example:

- **Human heights:** Heights of people in a large population follow a normal distribution, with most people being close to the average height and fewer people being extremely short or tall.
- **IQ scores:** IQ scores in a large population follow a normal distribution with an average of 100 and a standard deviation of 15.

3. Used in Statistical Inference and Hypothesis Testing

- Many **statistical tests** (e.g., Z-test, T-test, ANOVA) assume that the data follows a normal distribution.
- The **p-value** in hypothesis testing is often derived assuming normality.

Example:

A pharmaceutical company tests a new drug by comparing its effects on a **normally distributed** sample of patients to decide if the drug has a significant impact.

4. Basis for Standard Scores (Z-Scores & Percentiles)

- **Standard normal distribution** (mean = 0, standard deviation = 1) allows conversion of raw scores into **Z-scores**, which help in comparing different datasets.

Example:

A student scores **85** on a math test where the mean score is **75** and the standard deviation is **10**. The **Z-score** is:

$$Z = \frac{X - \mu}{\sigma} = \frac{85 - 75}{10} = 1$$

A **Z-score of 1** means the student scored **one standard deviation above the mean**, or better than about **84% of the class**.

5. Used in Machine Learning & Data Science

- Many **machine learning models** (e.g., linear regression, logistic regression) assume that the data follows a normal distribution.
- **Feature scaling** techniques like **standardization (Z-score normalization)** rely on normality.

Example:

In **stock market analysis**, financial analysts assume **daily stock returns** are normally distributed to predict risk and performance.

6. Important in Quality Control & Six Sigma

- Quality control processes assume **errors and deviations** follow a normal distribution.
- **Six Sigma methodology** uses normal distribution concepts to minimize product defects.

Example:

A factory produces light bulbs with an **average lifespan of 1000 hours** and a **standard deviation of 50 hours**. If bulbs last **less than 900 hours** or **more than 1100 hours**, they are considered defective.

Using the normal distribution:

- $P(X < 900)$ and $P(X > 1100)$ can be calculated using Z-scores to estimate the percentage of defective bulbs.

4. Example: Probability Calculation Using Normal Distribution

Scenario:

The **average weight of adult males** in a city is **75 kg** with a standard deviation of **5 kg**. What is the probability that a randomly selected male weighs **more than 80 kg**?

Step 1: Convert to Z-score

We use the Z-score formula:

$$\begin{aligned} Z &= \frac{X - \mu}{\sigma} \\ Z &= \frac{80 - 75}{5} \\ &= \frac{5}{5} \\ &= 1 \end{aligned}$$

Step 2: Find Probability from Z-Table

From the standard normal table:

$$P(Z < 1) = 0.8413$$

Since we need $P(X > 80)$:

$$P(X > 80) = 1 - P(Z < 1) = 1 - 0.8413 = 0.1587$$

Step 3: Interpret the Result

There is a **15.87% probability** that a randomly selected male weighs more than **80 kg**.